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Protocol

CustomDebugStringConvertible

A type with a customized textual representation suitable for debugging purposes.

iOS 8.0+ | iPadOS 8.0+ | Mac Catalyst 13.0+ | macOS 10.10+ | tvOS 9.0+ | visionOS 1.0+ | watchOS 2.0+

protocol CustomDebugStringConvertible

Overview

Swift provides a default debugging textual representation for any type. That default representation is used by the `String(reflecting:)` initializer and the debug `Print(_:)` function for types that don't provide their own. To customize that representation, make your type conform to the `CustomDebugStringConvertible` protocol.

Because the `String(reflecting:)` initializer works for instances of *any* type, returning an instance's `debugDescription` if the value passed conforms to `CustomDebugStringConvertible`, accessing a type's `debugDescription` property directly or using `CustomDebugStringConvertible` as a generic constraint is discouraged.

Note

Calling the `dump(_:_:_:_:)` function and printing in the debugger uses both `String(reflecting:)` and `Mirror(reflecting:)` to collect information about an instance. If you implement `CustomDebugStringConvertible` conformance for your custom type, you may want to consider providing a custom mirror by implementing `CustomReflectable` conformance, as well.

Conforming to the CustomDebugStringConvertible Protocol

Add `CustomDebugStringConvertible` conformance to your custom types by defining a `debugDescription` property.

For example, this custom `Point` struct uses the default representation supplied by the standard library:

```
struct Point {
    let x: Int, y: Int
}

let p = Point(x: 21, y: 30)
print(String(reflecting: p))
// Prints "Point(x: 21, y: 30)"
```